

Times Series Part II

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bookdown::render_book()

Overview

Topics for Week 12:

1. Unit Roots
2. Testing for Serial Correlation
3. Correcting for Serial Correlation

Chapter 1

Unit Roots

Lesson: when we have a unit root process or Time Series integrated of order one $I(1)$ we can use a first difference to transform the time series into a weakly dependent and often stationary process.

1.1 Example: Revisit Fertility

Set the time series

```
cd "/Users/Sam/Desktop/Econ 645/Data/Wooldridge"  
use fertil3.dta, clear  
tsset year
```

```
/Users/Sam/Desktop/Econ 645/Data/Wooldridge
```

```
time variable:  year, 1913 to 1984  
delta: 1 unit
```

Plot the Graph

```
twoway line gfr year, title("General Fertility Rate") ytitle("Children Born per 1,000 CBA Women")  
graph export "/Users/Sam/Desktop/Econ 645/Stata/week_11_i1.png", replace
```

Estimate autocorrelation or estimate $\hat{\rho}$

```
reg gfr l.gfr
```

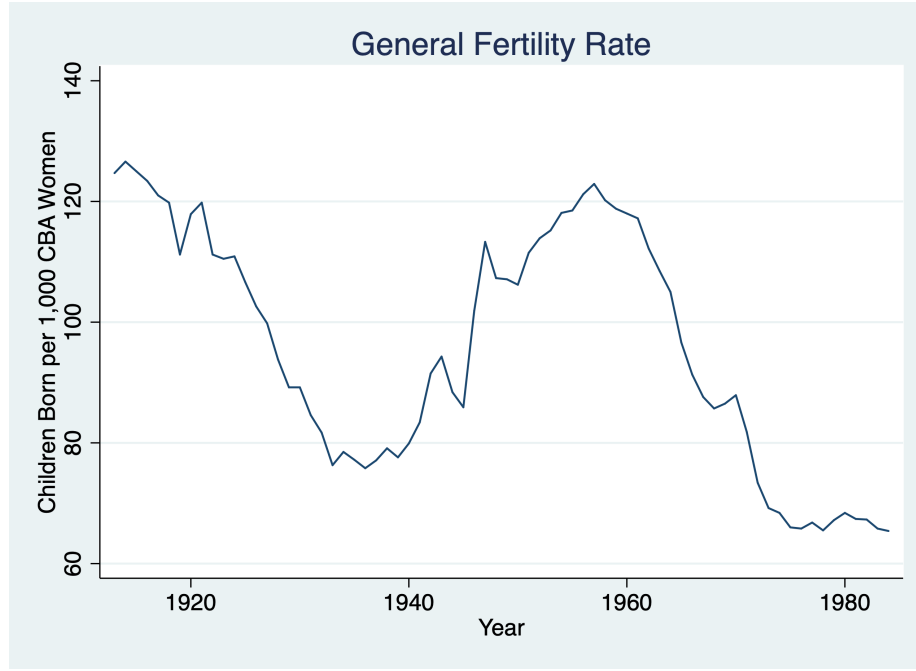


Figure 1.1: General Fertility Rate

Source	SS	df	MS	Number of obs	=	71
Model	25734.824	1	25734.824	F(1, 69)	=	1413.53
Residual	1256.21904	69	18.2060731	Prob > F	=	0.0000
Total	26991.043	70	385.586329	R-squared	=	0.9535
				Adj R-squared	=	0.9528
				Root MSE	=	4.2669

	gfr	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
	gfr					
	L1.	.9777202	.0260053	37.60	0.000	.925841 1.029599
	_cons	1.304937	2.548821	0.51	0.610	-3.779822 6.389695

Our result is rho-hat is 0.98, which is indicative of a unit root process. Under our TS assumptions our t statistics are invalid, but if we use a transformation using a first-difference process, we can relax those assumptions under the TSC assumptions. If our $I(1)$ becomes $I(0)$, then our TSC assumptions are potentially valid.

First Difference both dependent and independent

```
reg d.gfr d.pe if year < 1985
```

Source	SS	df	MS	Number of obs	=	71
				F(1, 69)	=	2.26
Model	40.3237206	1	40.3237206	Prob > F	=	0.1370
Residual	1229.25866	69	17.8153428	R-squared	=	0.0318
				Adj R-squared	=	0.0177
Total	1269.58238	70	18.1368911	Root MSE	=	4.2208

D.gfr	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
pe						
D1.	-.0426776	.0283672	-1.50	0.137	-.0992686	.0139134
_cons	-.7847796	.5020398	-1.56	0.123	-1.786322	.2167625

Our result is that the real value of the personal exemption is associated with a 0.043 decrease in fertility or a 23 dollar increase in the real value of the personal exemption is associated with a 1 child per capita (1,000 CBA Women). This is a bit unexpected, but still insignificant. Let's add some lags.

First difference with more than 1 lag

```
reg d.gfr d.pe d.pe_1 d.pe_2 if year < 1985
```

Source	SS	df	MS	Number of obs	=	69
				F(3, 65)	=	6.56
Model	293.259859	3	97.7532864	Prob > F	=	0.0006
Residual	968.199959	65	14.895384	R-squared	=	0.2325
				Adj R-squared	=	0.1971
Total	1261.45982	68	18.5508797	Root MSE	=	3.8595

D.gfr	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
pe						
D1.	-.0362021	.0267737	-1.35	0.181	-.089673	.0172687
pe_1						
D1.	-.0139706	.0275539	-0.51	0.614	-.0689997	.0410584

loutphr		1.639639	.0933471	17.56	0.000	1.450668	1.828611
t		-.01823	.0017482	-10.43	0.000	-.021769	-.0146909
_cons		-5.328454	.3744492	-14.23	0.000	-6.086487	-4.570421

We estimate that the elasticity is very large, where a 1% increase in labor productivity (output per hour) is associated with a 1.64% increase in hourly wage. Let's test for autocorrelation with accounting for the linear trend.

```
reg lhrwage l.lhrwage t
```

Source		SS	df	MS	Number of obs	=	40
	+				F(2, 37)	=	1595.37
Model		.921760671	2	.460880336	Prob > F	=	0.0000
Residual		.010688817	37	.000288887	R-squared	=	0.9885
	+				Adj R-squared	=	0.9879
Total		.932449488	39	.023908961	Root MSE	=	.017

lhrwage		Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
	+					
lhrwage						
L1.		.9842872	.0334331	29.44	0.000	.9165452 1.052029
t		-.0009039	.0004732	-1.91	0.064	-.0018626 .0000548
_cons		.0544169	.0413998	1.31	0.197	-.0294671 .1383008

We have some evidence for a unit root process, so we'll use the first-difference to transform the I(1) process into a I(0) process. We'll no longer need the time trend

```
twoway line lhrwage year, title("Natural Log of Hourly Wages") ytitle("Ln(Hourly Wage)") xtitle("Year")
graph export "/Users/Sam/Desktop/Econ 645/Stata/week_11_hrwage1.png", replace
```

```
twoway line d.lhrwage year, title("First-Difference in Ln(Hourly Wages)") ytitle("Delta Ln(Hourly Wage)") xtitle("Year")
graph export "/Users/Sam/Desktop/Econ 645/Stata/week_11_hrwage2.png", replace
```

We'll keep the trend based upon our prior graph

```
reg d.lhrwage d.loutphr t
```

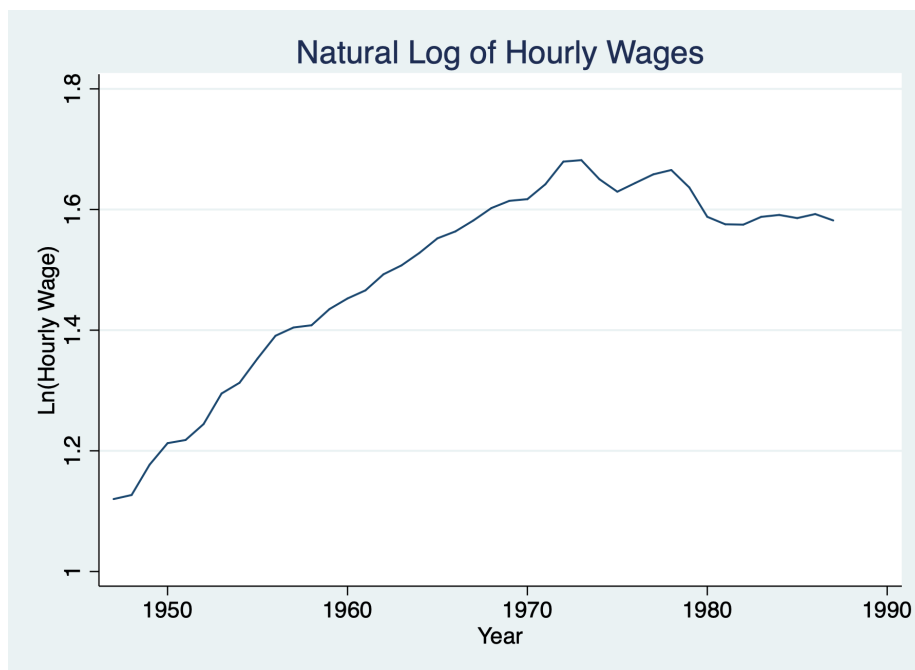


Figure 1.2: Natural Log of Hourly Wages

Source	SS	df	MS	Number of obs	=	40
Model	.008727211	2	.004363605	F(2, 37)	=	19.12
Residual	.008445792	37	.000228265	Prob > F	=	0.0000
Total	.017173003	39	.000440333	R-squared	=	0.5082
				Adj R-squared	=	0.4816
				Root MSE	=	.01511

D.lhrwage	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
loutphr						
D1.	.5511404	.1733698	3.18	0.003	.1998598	.9024209
t	-.0007637	.0002321	-3.29	0.002	-.0012339	-.0002935
_cons	.0176094	.0074784	2.35	0.024	.0024567	.0327622

Our transformed model shows that a 1% increase in output per hour is associated with a .55% increase in hourly wage.

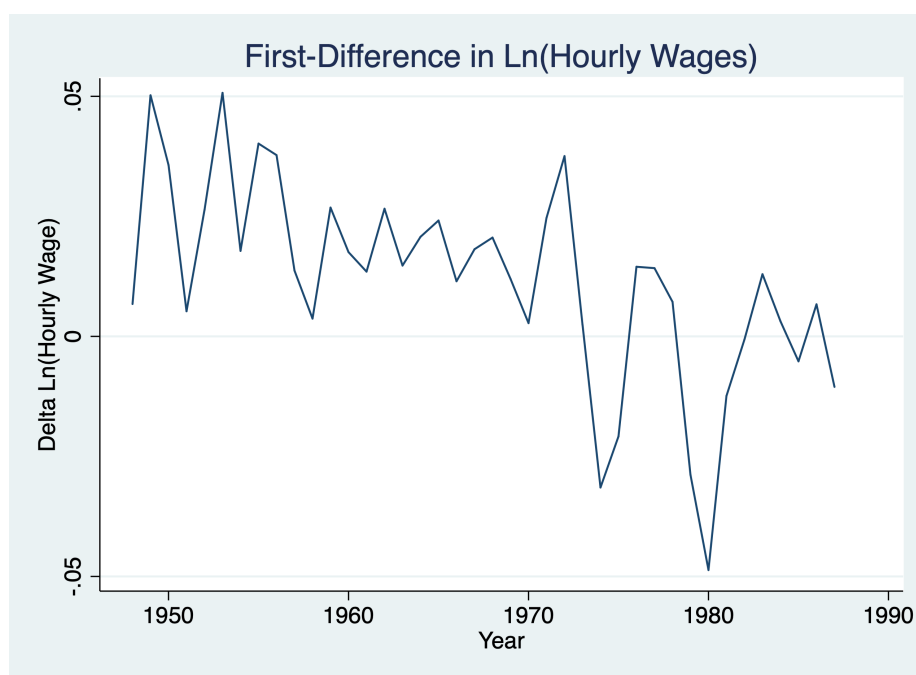


Figure 1.3: Difference in LN of Hourly Wage

Chapter 2

Dynamically Complete

Lesson: Is our model dynamically complete? Use theory/literature, but test, test, test.

Set Time Series

```
cd "/Users/Sam/Desktop/Econ 645/Data/Wooldridge"
use fertil3.dta
tsset year, yearly
```

```
/Users/Sam/Desktop/Econ 645/Data/Wooldridge
```

```
time variable: year, 1913 to 1984
delta: 1 year
```

We'll estimate a Finite Distributed Lag model for the first difference of general fertility onto the first difference of personal exemptions. If our model is dynamically complete, then no additional lags of gfr or pe are needed.

Our First-difference model with one lag of pe

```
reg d.gfr d.pe d.pe_1 if year < 1985
```

Source	SS	df	MS	Number of obs	=	70
Model	43.7985353	2	21.8992676	F(2, 67)	=	1.20
Residual	1218.19557	67	18.1820235	Prob > F	=	0.3063
				R-squared	=	0.0347
				Adj R-squared	=	0.0059
Total	1261.99411	69	18.2897697	Root MSE	=	4.264

D.gfr	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
pe						
D1.	-.0456497	.0294689	-1.55	0.126	-.1044699	.0131704
pe_1						
D1.	.0134149	.0295326	0.45	0.651	-.0455324	.0723623
_cons	-.8372962	.5117337	-1.64	0.106	-1.858721	.1841286

Is this dynamically complete? Let's add a second lag of pe

```
reg d.gfr d.pe d.pe_1 d.pe_2 if year < 1985
```

Source	SS	df	MS	Number of obs	=	69
Model	293.259859	3	97.7532864	F(3, 65)	=	6.56
Residual	968.199959	65	14.895384	Prob > F	=	0.0006
Total	1261.45982	68	18.5508797	R-squared	=	0.2325
				Adj R-squared	=	0.1971
				Root MSE	=	3.8595

D.gfr	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
pe						
D1.	-.0362021	.0267737	-1.35	0.181	-.089673	.0172687
pe_1						
D1.	-.0139706	.0275539	-0.51	0.614	-.0689997	.0410584
pe_2						
D1.	.1099896	.0268797	4.09	0.000	.0563071	.1636721
_cons	-.9636787	.4677599	-2.06	0.043	-1.89786	-.0294976

Add lag of d.gfr and see it wasn't dynamically complete

```
reg d.gfr d.gfr_1 d.pe d.pe_1 d.pe_2 if year < 1985
```

Source	SS	df	MS	Number of obs	=	69
				F(4, 64)	=	7.46

Model		401.286162	4	100.32154	Prob > F	=	0.0001
Residual		860.173657	64	13.4402134	R-squared	=	0.3181
-----+					Adj R-squared	=	0.2755
Total		1261.45982	68	18.5508797	Root MSE	=	3.6661

	D.gfr	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
gfr_1	D1.	.3002422	.1059034	2.84	0.006	.0886758 .5118086
pe	D1.	-.0454721	.0256417	-1.77	0.081	-.0966972 .005753
pe_1	D1.	.002064	.0267776	0.08	0.939	-.0514303 .0555584
pe_2	D1.	.1051346	.0255904	4.11	0.000	.054012 .1562572
_cons		-.7021594	.4537988	-1.55	0.127	-1.608727 .2044079

Chapter 3

Testing for Serial Correlation

3.1 Example: Revisit Phillip's Curve

Lesson: using our t-test, Durban Watson test, Breusch-Godfrey test, and Durban's alternative statistic to test for serial correlation

We'll use our t-test for serial correlation with strictly exogenous regressors under the assumption that our regressors are exogenous.

Set Time Series

```
cd "/Users/Sam/Desktop/Econ 645/Data/Wooldridge"  
use phillips.dta, clear  
tsset year, yearly
```

```
/Users/Sam/Desktop/Econ 645/Data/Wooldridge
```

```
time variable: year, 1948 to 2003  
delta: 1 year
```

We estimated a static Phillips Curve and an augmented Phillips Curve last week. We'll retest the models for serial correlation.

Static Model

Test for serial correlation in Static Model Noticeable serial correlation

```

*Static Model
reg inf unem if year < 1997

*Heteroskedastiity Test: Breusch-Godfrey Test
estat bgodfrey

*Serial Correlation Test: Simple AR(1) test
predict u, resid

reg u l.u, noconst

```

Source	SS	df	MS	Number of obs	=	49
Model	25.6369575	1	25.6369575	F(1, 47)	=	2.62
Residual	460.61979	47	9.80042107	Prob > F	=	0.1125
Total	486.256748	48	10.1303489	R-squared	=	0.0527
				Adj R-squared	=	0.0326
				Root MSE	=	3.1306

inf	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
unem	.4676257	.2891262	1.62	0.112	-.1140213	1.049273
_cons	1.42361	1.719015	0.83	0.412	-2.034602	4.881822

Breusch-Godfrey LM test for autocorrelation

lags(p)	chi2	df	Prob > chi2
1	18.472	1	0.0000

H0: no serial correlation

Source	SS	df	MS	Number of obs	=	55
Model	161.010846	1	161.010846	F(1, 54)	=	29.61
Residual	293.652171	54	5.43800317	Prob > F	=	0.0000
Total	454.663017	55	8.26660031	R-squared	=	0.3541
				Adj R-squared	=	0.3422
				Root MSE	=	2.332

u	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
u						

L1.		.5822453	.1070035	5.44	0.000	.3677161	.7967745
-----	--	----------	----------	------	-------	----------	----------

First difference

Test for serial correlation in Augmented Serial correlation is removed with the first-difference in inflation

```
*First Difference Model
reg d.inf unem if year < 1997
predict u2, resid
*Breusch-Godfrey Test
estat bgodfrey
```

Source		SS	df	MS	Number of obs	=	48
Model		33.3829996	1	33.3829996	F(1, 46)	=	5.56
Residual		276.305138	46	6.00663344	Prob > F	=	0.0227
					R-squared	=	0.1078
					Adj R-squared	=	0.0884
Total		309.688138	47	6.58910932	Root MSE	=	2.4508

D.inf		Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
unem		-.5425869	.2301559	-2.36	0.023	-1.005867 - .079307
_cons		3.030581	1.37681	2.20	0.033	.259206 5.801955

(1 missing value generated)

Breusch-Godfrey LM test for autocorrelation

lags(p)		chi2	df	Prob > chi2
1		0.062	1	0.8039

H0: no serial correlation

Test for Serial Correlation: Durban-Watson Test

```
estat dwatson
```

```
Durbin-Watson d-statistic( 2, 48) = 1.769648
```

$$d_L = 1.503$$

$$d_U = 1.585$$

$$1.769648 > 1.585$$

We fail to reject the null hypothesis.

Test for Serial Correlation: AR(1) Test

```
reg u2 l.u2, noconst
```

Source	SS	df	MS	Number of obs	=	54
Model	.267251656	1	.267251656	F(1, 53)	=	0.07
Residual	198.724905	53	3.74952651	Prob > F	=	0.7905
Total	198.992157	54	3.68503994	R-squared	=	0.0013
				Adj R-squared	=	-0.0175
				Root MSE	=	1.9364

u2	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
u2					
L1.	-.0308131	.1154154	-0.27	0.791	-.262307 .2006808

Test change in error terms for FD assumption on serial correlation so that the differences in errors are not correlated.

```
gen u2_1=l.u2
reg d.u2 d.u2_1, noconst
drop u2
```

(2 missing values generated)

Source	SS	df	MS	Number of obs	=	53
				F(1, 52)	=	2.01

Model		13.6239993	1	13.6239993	Prob > F	=	0.1623
Residual		352.532902	52	6.77947888	R-squared	=	0.0372
-----+					Adj R-squared	=	0.0187
Total		366.156901	53	6.90862078	Root MSE	=	2.6037

D.u2		Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
-----+							
u2_1							
D1.		-.1661051	.1171734	-1.42	0.162	-.4012307	.0690204

We fail to reject the null hypothesis that there is no serial correlation in the difference in error term.

FDL Model

Let's add a lag in inflation to change our static model into a FDL model. We can also use the estat dwatson or dwstat

```
reg inf l.inf unem if year < 1997
estat dwatson
*Or
dwstat
```

Source		SS	df	MS	Number of obs	=	48
-----+					F(2, 45)	=	19.72
Model		219.518187	2	109.759094	Prob > F	=	0.0000
Residual		250.471823	45	5.56604051	R-squared	=	0.4671
-----+					Adj R-squared	=	0.4434
Total		469.99001	47	9.99978744	Root MSE	=	2.3592

inf		Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
-----+							
inf							
L1.		.7278693	.1263167	5.76	0.000	.4734544	.9822841
unem		-.2443814	.26124	-0.94	0.355	-.7705457	.2817829
_cons		2.43082	1.354276	1.79	0.079	-.2968325	5.158473

Durbin-Watson d-statistic(3, 48) = 1.477467

```
Durbin-Watson d-statistic( 3, 48) = 1.477467
```

First, let's find our critical values: https://www3.nd.edu/~wevans1/econ30331/Durbin_Watson_tables.pdf

$$DW(3, 48) = 1.48$$

and

$$d_L \approx 1.4$$

and

$$d_U \approx 1.67$$

so our DW test is inconclusive. We'll look at Durbin-Watson and DW Tables again a bit later.

3.2 Example: Revisit Puerto Rican Employment and Minimum Wage

Lesson: We can use Durbin's alternative statistic to test for serial correlation if our regressors are not strictly exogenous.

Durban's alternative statistic

Set Time Series

```
cd "/Users/Sam/Desktop/Econ 645/Data/Wooldridge"
use prminwge, clear
tsset year
```

```
/Users/Sam/Desktop/Econ 645/Data/Wooldridge
```

```
time variable: year, 1950 to 1987
delta: 1 unit
```

We'll rerun our static model and regress the natural log of the employment-population ratio onto the natural log of the importance of minimum wage, the natural log of Puerto Rican Gross National Product, the natural log of US GNP, and a time trend.

Durbin-Watson Test

3.2. EXAMPLE: REVISIT PUERTO RICAN EMPLOYMENT AND MINIMUM WAGE25

```
reg lprepop lmincov lprgnp lusgnp t if year < 1988
predict u, resid
```

Source	SS	df	MS	Number of obs	=	38
				F(4, 33)	=	66.23
Model	.284430187	4	.071107547	Prob > F	=	0.0000
Residual	.035428331	33	.001073586	R-squared	=	0.8892
				Adj R-squared	=	0.8758
Total	.319858518	37	.008644825	Root MSE	=	.03277

	lprepop	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
	lmincov	-.2122612	.0401524	-5.29	0.000	-.2939518 -.1305706
	lprgnp	.2852386	.0804922	3.54	0.001	.121476 .4490012
	lusgnp	.4860463	.2219829	2.19	0.036	.0344188 .9376739
	t	-.0266633	.0046267	-5.76	0.000	-.0360765 -.0172502
	_cons	-6.663432	1.257831	-5.30	0.000	-9.222508 -4.104356

Durbin's Alternative Statistics

Test Serial Correlation with Durbin's alternative statistic, which is valid when we don't have strictly exogenous regressors.

```
reg u l.u lmincov lprgnp lusgnp t, noconst
```

Source	SS	df	MS	Number of obs	=	37
				F(5, 32)	=	1.92
Model	.007182064	5	.001436413	Prob > F	=	0.1191
Residual	.023990478	32	.000749702	R-squared	=	0.2304
				Adj R-squared	=	0.1101
Total	.031172542	37	.000842501	Root MSE	=	.02738

	u	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
	u					
	L1.	.4757945	.1653074	2.88	0.007	.1390742 .8125147
	lmincov	.041425	.0346345	1.20	0.240	-.0291232 .1119732
	lprgnp	-.0522061	.0615544	-0.85	0.403	-.1775883 .0731762
	lusgnp	.0606233	.0644769	0.94	0.354	-.070712 .1919585

```
t | -.0005264 .0015194 -0.35 0.731 -.0036213 .0025685
```

We can see that we reject the null hypothesis, since our $\hat{rho} = 0.48$ and statistically significant.

3.3 Test AR(3) serial correlation

Lesson: We can test for serial correlation among prior periods beside AR(1).

Set Time Series

```
cd "/Users/Sam/Desktop/Econ 645/Data/Wooldridge"
use barium.dta, clear
tsset t, monthly
```

```
/Users/Sam/Desktop/Econ 645/Data/Wooldridge
```

```
time variable: t, 1960m2 to 1970m12
delta: 1 month
```

We'll look at our barium chloride model again to see if ITC complaints affect behavior of foreign exporters. We may have higher order of serial correlation since we are using monthly data.

```
reg lchnimp lchempi lgas lrtwex befile6 affile6 afdec6
predict u, resid
```

Source	SS	df	MS	Number of obs	=	131
-----+-----				F(6, 124)	=	9.06
Model	19.4051607	6	3.23419346	Prob > F	=	0.0000
Residual	44.2470875	124	.356831351	R-squared	=	0.3049
-----+-----				Adj R-squared	=	0.2712
Total	63.6522483	130	.489632679	Root MSE	=	.59735

lchnimp	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
-----+-----					
lchempi	3.117193	.4792021	6.50	0.000	2.168718 4.065668
lgas	.1963504	.9066172	0.22	0.829	-1.598099 1.9908
lrtwex	.9830183	.4001537	2.46	0.015	.1910022 1.775034
befile6	.0595739	.2609699	0.23	0.820	-.4569585 .5761064
affile6	-.0324064	.2642973	-0.12	0.903	-.5555249 .490712

afdec6		-.565245	.2858352	-1.98	0.050	-1.130993	.0005028
_cons		-17.803	21.04537	-0.85	0.399	-59.45769	23.85169

Test for $H_0 : p_1 = p_2 = p_3 = 0$ jointly with an F-test.

```
reg u l.u l2.u l3.u lchempi lgas lrtwex befile6 affile6 afdec6, noconst
test l.u l2.u l3.u
```

Source		SS	df	MS	Number of obs	=	128
Model		4.8836741	9	.542630456	F(9, 119)	=	1.67
Residual		38.5511192	119	.323958985	Prob > F	=	0.1024
Total		43.4347933	128	.339334323	R-squared	=	0.1124
					Adj R-squared	=	0.0453
					Root MSE	=	.56917

u		Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
u						
L1.		.215573	.0910635	2.37	0.020	.0352581 .3958878
L2.		.1296804	.0917465	1.41	0.160	-.0519868 .3113477
L3.		.1209405	.0906816	1.33	0.185	-.0586181 .3004992
lchempi		-.105758	.4679356	-0.23	0.822	-1.032317 .8208012
lgas		.0125667	.1187882	0.11	0.916	-.2226459 .2477792
lrtwex		.0518992	.3452261	0.15	0.881	-.631683 .7354815
beffile6		-.0730016	.249775	-0.29	0.771	-.567581 .4215779
affile6		-.122359	.2541453	-0.48	0.631	-.6255919 .380874
afdec6		-.0694522	.2737453	-0.25	0.800	-.6114953 .4725909

```
( 1)  L.u = 0
( 2)  L2.u = 0
( 3)  L3.u = 0
```

```
F( 3, 119) = 4.98
Prob > F = 0.0027
```

Our reject the null hypothesis that all of the lags are equal to zero using an F-test.

However, when we test our last two lags, they are jointly insignificant.

```
test l2.u l3.u  
drop u
```

```
( 1)  L2.u = 0  
( 2)  L3.u = 0
```

```
      F( 2, 119) =    2.42  
      Prob > F =    0.0930
```

We have evidence of at least serial correlation to the order of one.

Chapter 4

Correct for Serial Correlation

4.1 Prais-Winsten Estimation in the Event Study

Lesson: We'll account for serial correlation with a FGLS Prais-Winsten model.

Set Time Series

```
cd "/Users/Sam/Desktop/Econ 645/Data/Wooldridge"  
use barium.dta, clear  
tsset t, monthly
```

```
/Users/Sam/Desktop/Econ 645/Data/Wooldridge
```

```
time variable: t, 1960m2 to 1970m12  
delta: 1 month
```

OLS

```
est clear  
eststo OLS: reg lchnimp lchempi lgas lrtwex befile6 affile6 afdec6  
estat dwstat
```

Source	SS	df	MS	Number of obs	=	131
--------	----	----	----	---------------	---	-----

-----+-----				F(6, 124)	=	9.06
Model		19.4051607	6	3.23419346	Prob > F	= 0.0000
Residual		44.2470875	124	.356831351	R-squared	= 0.3049
-----+-----				Adj R-squared	=	0.2712
Total		63.6522483	130	.489632679	Root MSE	= .59735

-----+-----						
lchnimp		Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
-----+-----						
lchempi		3.117193	.4792021	6.50	0.000	2.168718 4.065668
lgas		.1963504	.9066172	0.22	0.829	-1.598099 1.9908
lrtwex		.9830183	.4001537	2.46	0.015	.1910022 1.775034
befile6		.0595739	.2609699	0.23	0.820	-.4569585 .5761064
affile6		-.0324064	.2642973	-0.12	0.903	-.5555249 .490712
afdec6		-.565245	.2858352	-1.98	0.050	-1.130993 .0005028
_cons		-17.803	21.04537	-0.85	0.399	-59.45769 23.85169
-----+-----						

Durbin-Watson d-statistic(7, 131) = 1.458414

Prais-Winsten

```
eststo Prais: prais lchnimp lchempi lgas lrtwex befile6 affile6 afdec6
```

```
Iteration 0: rho = 0.0000
Iteration 1: rho = 0.2708
Iteration 2: rho = 0.2910
Iteration 3: rho = 0.2930
Iteration 4: rho = 0.2932
Iteration 5: rho = 0.2932
Iteration 6: rho = 0.2932
Iteration 7: rho = 0.2932
```

Prais-Winsten AR(1) regression -- iterated estimates

Source		SS	df	MS	Number of obs	=	131
-----+-----					F(6, 124)	=	5.24
Model		10.3252323	6	1.72087205	Prob > F	=	0.0001
Residual		40.7593886	124	.328704747	R-squared	=	0.2021
-----+-----					Adj R-squared	=	0.1635
Total		51.0846209	130	.392958622	Root MSE	=	.57333

lchnimp	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
lchempi	2.940949	.6328402	4.65	0.000	1.688381	4.193517
lgas	1.04638	.9773356	1.07	0.286	-.8880406	2.980801
lrtwex	1.132791	.5066578	2.24	0.027	.1299738	2.135609
befile6	-.0164787	.3193802	-0.05	0.959	-.6486216	.6156641
affile6	-.0331563	.3218101	-0.10	0.918	-.6701086	.603796
afdec6	-.5768122	.3419865	-1.69	0.094	-1.253699	.1000748
_cons	-37.0777	22.7783	-1.63	0.106	-82.16235	8.006941
rho	.293217					
Durbin-Watson statistic (original)			1.458414			
Durbin-Watson statistic (transformed)			2.087181			

First, let's find our critical values: https://www3.nd.edu/~wevans1/econ30331/Durbin_Watson_tables.pdf.
For the OLS Estimator

$$DW(6, 131) = 1.458414$$

For the Prasi-Winsten Estimator

$$DW(6, 131) = 2.087181$$

and

$$d_L \approx 1.60$$

and

$$d_U \approx 1.81$$

We reject the null hypothesis for the OLS estimator, but we fail to reject the null hypothesis for the Prais-Winsten Estimator.

Our beta-hats in Prais-Winsten Estimator are similar to our OLS estimates, but our PW standard errors account for the serial correlation. Our OLS standard errors usually understate the actual sampling variation in the OLS estimators and should be treated with suspicion.

```
esttab, mtitle se stat(N rho)
```

	(1) OLS	(2) Prais
lchempi	3.117*** (0.479)	2.941*** (0.633)

lgas	0.196 (0.907)	1.046 (0.977)
lrtwex	0.983* (0.400)	1.133* (0.507)
befile6	0.0596 (0.261)	-0.0165 (0.319)
affile6	-0.0324 (0.264)	-0.0332 (0.322)
afdec6	-0.565 (0.286)	-0.577 (0.342)
_cons	-17.80 (21.05)	-37.08 (22.78)

N	131	131
rho		0.293

Standard errors in parentheses

* p<0.05, ** p<0.01, *** p<0.001

4.2 Inflation and Prais

Lesson: our PW estimator might not be unbiased or consistent if our assumptions fail.

We'll show another example comparing OLS and Prais-Winsten estimators. We'll look at the static Phillips curve, which we know has serial correlation.

Set the Time Series

```
cd "/Users/Sam/Desktop/Econ 645/Data/Wooldridge"
use phillips.dta, clear
tsset year
est clear
```

```
/Users/Sam/Desktop/Econ 645/Data/Wooldridge
```

```
time variable:  year, 1948 to 2003
               delta:  1 unit
```

OLS

```
quietly eststo OLS: reg inf unem if year < 1997
```

Prais-Winsten

```
quietly eststo PW: prais inf unem if year < 1997
```

Prais-Winsten with Cochrane-Orcutt Transformation

```
quietly eststo PW_CO: prais inf unem if year < 1997, corc
```

First Difference

```
quietly eststo FD: reg d.inf d.unem if year < 1997
```

Compare

```
esttab, mtitle se stat(N rho)
```

	(1) OLS	(2) PW	(3) PW_CO	(4) FD
unem	0.468 (0.289)	-0.716* (0.313)	-0.665* (0.320)	
D.unem				-0.842* (0.314)
_cons	1.424 (1.719)	8.296*** (2.231)	7.583** (2.381)	-0.0782 (0.348)
N	49	49	48	48
rho		0.781	0.774	

Standard errors in parentheses

* p<0.05, ** p<0.01, *** p<0.001

The OLS and PW estimators might give very different estimators if our assumptions of strict exogeneity and the a $Cov(x_t + 1) + x_t - 1, u_t) = 0$ fail. If that is the case then using OLS with a first-difference might be a better method.

We can see that the difference is quite notable. With our OLS estimators, the tradeoff between inflation and employment is non-existent, but with the PW estimator our estimate of beta 1 is closer to our first-difference estimate of beta 1.

It is possible that there is no relationship with the static model, but the change in inflation and the change in unemployment are negatively related.

4.3 Differencing and Serial Correlation

4.3.1 Inflation and Deficits on Interest Rates

Lesson: First-differencing is a potential method to eliminate serial correlation.

Set the Time Series

```
cd "/Users/Sam/Desktop/Econ 645/Data/Wooldridge"
use intdef.dta, clear
tsset year
```

```
/Users/Sam/Desktop/Econ 645/Data/Wooldridge
```

```
time variable:  year, 1948 to 2003
               delta:  1 unit
```

We'll look at the relationship between interest rates from the 3-month T-bill rate (i3) and inflation rate (inf) and deficits as a percentage of GDP.

Static Model

Calculate a static model and test for serial correlation

```
reg i3 inf def if year < 2004
estat dwstat
```

Source		SS		df		MS		Number of obs	=	56
-----+-----										
Model		272.420338		2		136.210169		F(2, 53)	=	40.09
Residual		180.054275		53		3.39725047		Prob > F	=	0.0000
								R-squared	=	0.6021

-----+-----				Adj R-squared	=	0.5871
Total		452.474612	55	8.22681113	Root MSE	= 1.8432

-----+-----						
i3		Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
-----+-----						
inf		.6058659	.0821348	7.38	0.000	.4411243 .7706074
def		.5130579	.1183841	4.33	0.000	.2756095 .7505062
_cons		1.733266	.431967	4.01	0.000	.8668497 2.599682
-----+-----						

Durbin-Watson d-statistic(3, 56) = .7161527

Critical values $d(3, 56) = .716$, at the 5% level $dL = 1.452$, $dU = 1.681$ $dwstat < dL$ so we reject the null hypothesis

```
predict u, resid
reg u l.u, noconst
drop u
```

Source		SS	df	MS	Number of obs	=	55
-----+-----					F(1, 54)	=	32.83
Model		64.1034163	1	64.1034163	Prob > F	=	0.0000
Residual		105.448636	54	1.95275252	R-squared	=	0.3781
-----+-----					Adj R-squared	=	0.3666
Total		169.552053	55	3.08276459	Root MSE	=	1.3974

-----+-----						
u		Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
-----+-----						
u						
L1.		.6228816	.1087148	5.73	0.000	.4049216 .8408415
-----+-----						

We find that we have notable serial correlation in the error term that is statistically significant. We'll run a first-difference model next.

First differencing

Run a first-difference and test for serial correlation

```
reg d.i3 d.inf d.def if year < 2004
dwstat
```

Source	SS	df	MS	Number of obs	=	55
Model	17.8058164	2	8.90290822	F(2, 52)	=	5.57
Residual	83.1753707	52	1.59952636	Prob > F	=	0.0065
				R-squared	=	0.1763
				Adj R-squared	=	0.1446
Total	100.981187	54	1.87002198	Root MSE	=	1.2647

D.i3	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
inf					
D1.	.1494892	.0921555	1.62	0.111	-.0354343 .3344127
def					
D1.	-.1813151	.1476825	-1.23	0.225	-.4776618 .1150315
_cons	.0417738	.1713874	0.24	0.808	-.3021401 .3856877

Durbin-Watson d-statistic(3, 55) = 1.796365

Critical values $d(3,56) = 1.796$, at the 5% level $dL = 1.452, dU = 1.681$
 $dwstat > dU$ so we fail to reject the null hypothesis

```
predict u, resid
reg u l.u, noconst
```

(1 missing value generated)

Source	SS	df	MS	Number of obs	=	54
Model	.427058269	1	.427058269	F(1, 53)	=	0.29
Residual	77.8807184	53	1.46944752	Prob > F	=	0.5921
				R-squared	=	0.0055
				Adj R-squared	=	-0.0133
Total	78.3077766	54	1.45014401	Root MSE	=	1.2122

u	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
u					

L1.		.0717246	.133046	0.54	0.592	-.1951318	.3385811
-----	--	----------	---------	------	-------	-----------	----------

The serial correlation is no longer a problem with our first difference model, since first differencing can transform a unit root process $I(1)$ into a $I(0)$ process.

4.4 Serial Correlation Robust Standard Errors

Lesson: We can use the Newey estimator for estimates with Heteroskedastic and Autocorrelation Consistent (HAC) standard errors.

We'll revisit our Puerto Rican examination of the impact of minimum wage importance. We'll use Heteroskedastic and Autocorrelation Consistent (HAC) standard errors, and we'll compare OLS, OLS with HAC standard errors, and Prais-Winsten estimates. We'll allow a maximum lag order of autocorrelation up to 2 with the lag(2) option.

Set Time Series

```
cd "/Users/Sam/Desktop/Econ 645/Data/Wooldridge"
use prminwge.dta, clear
tsset year, yearly
```

```
/Users/Sam/Desktop/Econ 645/Data/Wooldridge
```

```
time variable: year, 1950 to 1987
delta: 1 year
```

OLS

```
est clear
eststo OLS: quietly reg lprepop lmincov lprgnp lusgnp t if year < 1988
```

Newey-West Estimator Order of 1

Newey with lag order 1

```
eststo Newey1: quietly newey lprepop lmincov lprgnp lusgnp t if year < 1988, lag(1)
```

Newey-West Estimator Order of 2

```
eststo Newey2: quietly newey lprepop lmincov lprgnp lusgnp t if year < 1988, lag(2)
```

Prais-Winsten

```
eststo PW: quietly prais lprepop lmincov lprgnp lusgnp t if year < 1988
```

With t-stats below estimated parameters

```
esttab, mtitle scalars(F N)
```

	(1) OLS	(2) Newey1	(3) Newey2	(4) PW
lmincov	-0.212*** (-5.29)	-0.212*** (-4.67)	-0.212*** (-4.64)	-0.148** (-3.22)
lprgnp	0.285** (3.54)	0.285** (2.85)	0.285** (2.86)	0.251* (2.16)
lusgnp	0.486* (2.19)	0.486 (1.79)	0.486 (1.74)	0.256 (1.10)
t	-0.0267*** (-5.76)	-0.0267*** (-4.86)	-0.0267*** (-4.63)	-0.0205** (-3.50)
_cons	-6.663*** (-5.30)	-6.663*** (-4.52)	-6.663*** (-4.34)	-4.653** (-3.38)
N	38	38	38	38
F	66.23	40.98	37.84	24.87

t statistics in parentheses

* p<0.05, ** p<0.01, *** p<0.001

With standard errors below the estimated parameters

```
esttab, mtitle se scalars(F N)
```

	(1) OLS	(2) Newey1	(3) Newey2	(4) PW
lmincov	-0.212*** (0.0402)	-0.212*** (0.0455)	-0.212*** (0.0457)	-0.148** (0.0458)
lprgnp	0.285** (0.0805)	0.285** (0.100)	0.285** (0.0996)	0.251* (0.116)
lusgnp	0.486* (0.222)	0.486 (0.272)	0.486 (0.279)	0.256 (0.232)
t	-0.0267*** (0.00463)	-0.0267*** (0.00549)	-0.0267*** (0.00576)	-0.0205** (0.00586)
_cons	-6.663*** (1.258)	-6.663*** (1.475)	-6.663*** (1.536)	-4.653** (1.376)
N	38	38	38	38
F	66.23	40.98	37.84	24.87

Standard errors in parentheses

* p<0.05, ** p<0.01, *** p<0.001

Notice how our t-stat have fallen and our standard errors have risen for our Newey1 and Newey2 compared to OLS. Also, notice how the Prais-Winsten beta-hat on minimum coverage is closer to 0, then OLS or OLS with HAC standard errors.

4.5 Testing for Heteroskedasticity in Time Series

We'll revisit our test of the Efficient Market Hypothesis with stock return data. We can test to see if our assumption of homoskedasticity is valid with our time series. This is a separate test than the test for serial correlation.

Set the time series

```
cd "/Users/Sam/Desktop/Econ 645/Data/Wooldridge"
use nyse.dta, clear
tsset t, weekly
```

```
/Users/Sam/Desktop/Econ 645/Data/Wooldridge
```

```
time variable:  t, 1960w2 to 1973w16
               delta:  1 week
```

OLS

We'll test heteroskedasticity with estat hettest

```
reg return l.return
estat hettest
```

Source	SS	df	MS	Number of obs	=	689
Model	10.6866231	1	10.6866231	F(1, 687)	=	2.40
Residual	3059.73817	687	4.45376735	Prob > F	=	0.1218
Total	3070.42479	688	4.46282673	R-squared	=	0.0035
				Adj R-squared	=	0.0020
				Root MSE	=	2.1104

return	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
return						
L1.	.0588984	.0380231	1.55	0.122	-.0157569	.1335538
_cons	.179634	.0807419	2.22	0.026	.0211034	.3381646

Breusch-Pagan / Cook-Weisberg test for heteroskedasticity

Ho: Constant variance

Variables: fitted values of return

chi2(1) = 95.22

Prob > chi2 = 0.0000

We'll manually calculate the Breusch-Pagan test

```
predict u, residu
gen u2=u^2
reg u2 l.return
```

(2 missing values generated)

(2 missing values generated)

Source	SS	df	MS	Number of obs	=	689
				F(1, 687)	=	30.05
Model	3755.56877	1	3755.56877	Prob > F	=	0.0000
Residual	85846.2961	687	124.958219	R-squared	=	0.0419
				Adj R-squared	=	0.0405
Total	89601.8649	688	130.235269	Root MSE	=	11.178

u2	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
return						
L1.	-1.104133	.2014029	-5.48	0.000	-1.499572	-.7086934
_cons	4.656501	.4276789	10.89	0.000	3.816786	5.496216

Test for serial correlation

```
reg u l.u, noconst
```

Source	SS	df	MS	Number of obs	=	688
				F(1, 687)	=	0.00
Model	.00603936	1	.00603936	Prob > F	=	0.9706
Residual	3059.08227	687	4.45281262	R-squared	=	0.0000
				Adj R-squared	=	-0.0015
Total	3059.08831	688	4.44634929	Root MSE	=	2.1102

u	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
u						
L1.	.001405	.0381496	0.04	0.971	-.0734987	.0763087

We can see we reject the null hypothesis that the variance of the error term is constant, and heteroskedasticity is a problem.

Newey-West Estimator

We can use robust, or HAC standard errors, but since serial correlation is not a problem, then we don't need newey estimator.

```
newey return l.return, lag(1)
```

```
Regression with Newey-West standard errors      Number of obs      =      689
maximum lag: 1                                F( 1,      687) =      0.74
                                              Prob > F          =      0.3895
```

		Newey-West					
return		Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	

return							
L1.		.0588984	.0684001	0.86	0.389	-.0754	.1931968
_cons		.179634	.0855634	2.10	0.036	.0116369	.3476311

Expected value does not depend on past returns, but the variance of the error term is not constant, and needs to be corrected.

Mitchell, N.M. (2020). *Data Management Using Stata: A Practical Handbook*. (2nd ed.). Stata Press.

UCLA Advanced Research Computing. (2025). *How can I access information stored after I run a command in Stata*. Retrieved from <https://stats.oarc.ucla.edu/stata/faq/how-can-i-access-information-stored-after-i-run-a-command-in-stata-returned-results/>

Wooldridge (2009). *Introductory Econometrics: A Modern Approach*. (4th ed.). South-Western Cengage Learning.